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Research and application of hyperledger fabric based on VRF on electronic order

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Abstract

With the rapid development of information digitization, a large number of electronic orders have also brought a series of problems such as data security and trust. In the face of the ever-increasing demand for electronic data storage, the traditional and conventional storage methods have gradually exposed many problems such as extremely high cost, low efficiency, and difficulty in acceptance. It has become an urgent need to ensure the data security and credibility of electronic orders. solved problem. Aiming at how to ensure the authority and security of electronic documents, through the study of the Hyperledger Fabric consensus mechanism, and at the same time, in order to solve the security risks and processing bottlenecks caused by the use of fixed endorsement nodes in the hyperledger structure to handle all exchanges, the use of A non-interactive, verifiable random endorsement scheme. On the basis of the consensus mechanism of "endorsement-sorting-verification", by introducing a candidate set of endorsing nodes, using a verifiable random function to randomly select endorsing nodes and endorsing each order, a non-interactive verifiable random selection of endorsing nodes is realized And parallel endorsement process. Analysis and experiments show that the optimized consensus mechanism has higher security and faster transaction processing speed.

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1. Introduction

One of the core problems of blockchain technology is to reach consensus among nodes in a decentralized, unreliable distributed system. Consensus simply means that the nodes in a distributed system have the same view of a value or a state. Consensus algorithm is the core technology of blockchain - consensus mechanism. "Blockchain + electronic order" has gradually become a research and development direction of electronic order experts at home and abroad. Based on the open, transparent and non-tamper able data of blockchain technology, it is applied to electronic order management of materials and other aspects, which will contribute to the transformation and development of the field of materials. Aiming at the electronic order business scenario, in order to solve the problem of low transaction throughput and single point of failure of centralized storage, this paper uses Hyperledger as the underlying blockchain technology platform, and uses VRF to ensure the security of endorsing nodes. Increase the trading volume of electronic orders.

2. Technical background

2.1. Introduction to Blockchain

The blockchain is based on the chain block data structure shown in Figure 1. A block contains two parts: the block header and the block body 1. The block header contains the hash value of the previous block, the Merkle root, the timestamp, the version number and other information. Among them, the Merkle root is generated from the total hash value of all transaction information in the block body. The hash value of the previous block is used as the connection index between the blocks, and the blocks are connected in chronological order to form a main chain. The timestamp is used to determine the time for each generated block to ensure that the block cannot be forged. Other information may contain random number, target hash value and other data according to the different consensus mechanism 2. The transaction information in the block body is stored by the Merkle tree 3. This storage structure can ensure that the root value of Merkle changes after the transaction information is tampered with, and that the transaction information cannot be tampered with. The bottom layer of the block chain relies on the P2P network 4, each node holds a copy of the ledger, and all copies are updated simultaneously. Unless you have control over most of the nodes in the network, it makes no sense to change the data of a single node in the network.

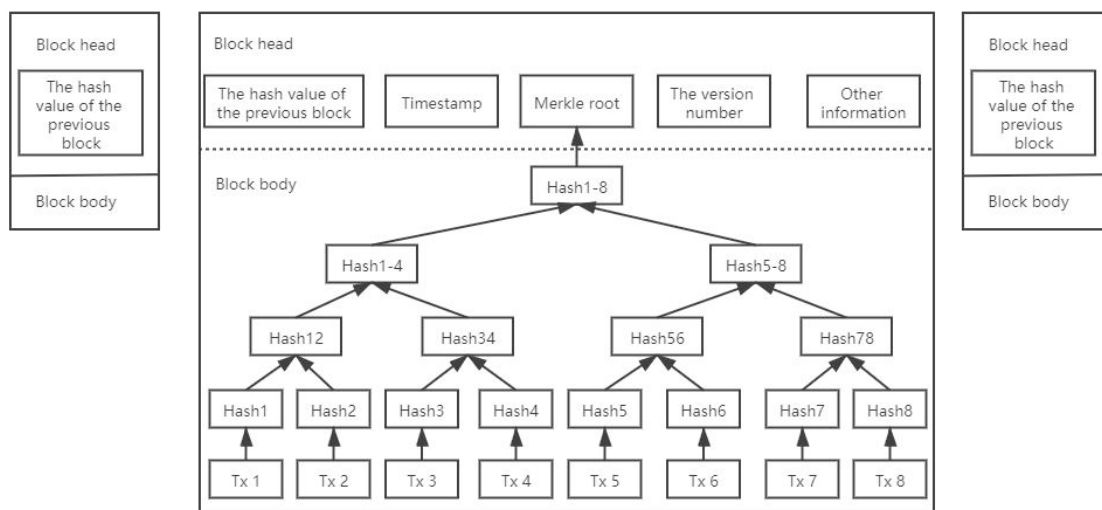


Fig. 1. Blockchain data structure.

2.2. Verifiable random function

Verifiable Random Function (VRF) ⁵ was proposed by Silvio Micali et al. in 1999. VRF is essentially a kind of pseudo-random function with verification function. For a specific input x and input The private key SK and VRF will output a random number v and a proof. The verifier can verify whether the random number is generated by the input through the three parts of the output random number, the proof and the input. This process does not need to expose the input. The private key is therefore safe.

VRF needs to meet three properties: verifiability, determinism and randomness:

- Verifiability means that through proof, it can be verified that v is the output corresponding to SK and x .
- Determinism means that when SK and x are unchanged, the output v is also unchanged.
- Randomness means that the output v of VRF and a random number are indistinguishable from the adversary without a proof.

3. The consensus mechanism based on VRF (Verifiable Random Functions)

Hyperledger Fabric's consensus model, which decouples application layer trust from underlying consensus protocol, is very good for building consensus mechanism based on business logic, so it has been widely used in practical systems. The relevant research work of Hyperledger Fabric consensus mechanism, under the assumption of high trust in endorsement nodes, mainly focuses on the improvement of sorting process, that is, the optimization of transaction order consistency ⁶. In fact, since the security of the transaction is highly dependent on the endorsement node with fixed identity, once the endorsement node is maliciously attacked, the correctness of the final accounting data cannot be guaranteed. Due to the open network environment faced by the public chain system, the design of its consensus mechanism takes security into full consideration, and the method of random extraction of consensus nodes is often used to make it difficult for the adversary to choose the target for attack, thus increasing the difficulty of attack. Therefore, aiming at the security problem of consensus mechanism in alliance chain, this paper introduces non-interactive verifiable random node extraction method on the basis of "endorsement - sequencing - verification" consensus model, and proposes a Hyperledger Fabric consensus mechanism

Optimization scheme:

- 1) Introducing the endorsement node candidate set, using the Random output feature of Verifiable Random Functions (VRF) to extract endorsement nodes under non-interactive mode, reducing the centralization of endorsement nodes and increasing the anti-attack capability of the system;
- 2) The transaction client USES the verification feature of verifiable random function to complete the identity authentication of endorsement nodes, so as to ensure the authenticity and verifiability of the identity of endorsement nodes in the process of random node extraction;
- 3) The original set of endorsement nodes should be used to endorse all the transactions, and multiple groups of endorsement nodes should be improved to endorse the transactions. The method of parallel processing should be adopted to improve the transaction processing speed.

3.1. Fabric consensus mechanism after using VRF

The optimized consensus mechanism is shown in Figure 2:

The client generates a proposal $proposal < req, s >_{sig}$, where req is the transaction data, including the *chaincode* and its parameters that it wants to call. s is the random value selected by the client as the seed of the node identity extraction algorithm. The client signs the *proposal* and sends it to the candidate set of endorsing nodes. After the transaction is successfully sent, the client will start a timer;

After each candidate endorsement node receives the *proposal* from the client, it first verifies the integrity of the proposal according to the signature, and terminates the transaction if the verification fails. The candidate endorsement node executes the endorsement node identity extraction algorithm $(r, proof, result) = VRF_Result(s, SK)$, And judge whether you are an endorsing node based on the *result*.

If you determine that you are an endorsing node, execute the proposal and generate the read-write set rw_set and the endorsement result edm . Then generate the proposal response message:
 $proposal_response < rw_set, edm, (r, proof) >_{sig}$.

The endorsement node extraction algorithm flow is as follows Show:

1) Generate random numbers and their proofs according to the input

$(r, proof) = F(s, SK)$

2) Return the lottery result

Calculate the extraction result according to r and compare it with the threshold λ , where $hash()$ is the password hash algorithm, $hashlen$ is the output length of the hash algorithm, if $\frac{hash(r)}{2^{hashlen}} > \lambda$, return $(r, proof, yes)$; otherwise return $(r, proof, no)$.

(4) During the period before the timer expires, the client continues to collect $proposal_response$ from different endorsing nodes and verifies the integrity of the $proposal_response$ according to the signature. If the verification fails, the transaction is terminated. Use the endorsing node identity verification algorithm $VRF_Verify()$ Verify whether the node is a legal endorsement node. If it is not, discard the endorsement result. In a legal $proposal_response$, if most (more than half) of the read and write sets are consistent, then generate a transaction $tx < r_w_set$ based on these endorsement results $\{edm\}^k_{sig}$. Where $\{edm\}^k$ represents the signature from k legal endorsing nodes. The client sends the transaction tx to the ordering node after signing the transaction. The authentication algorithm flow of the endorsing node is as follows:

1) Verify the legality of random numbers $bool = V(s, r, PK)$

2) Return the verification result

If the random number is legal and meets the threshold condition, it is deemed to be an endorsement node. If $bool \& \frac{hash(r)}{2^{hashlen}} > \lambda$, return yes ; otherwise return no .

It should be noted that since the identity extraction of endorsement nodes is a probabilistic algorithm, there may be a case where a transaction does not have a corresponding endorsement node, so it is necessary to set an appropriate threshold λ to reduce the probability of this situation. At the same time, if the $proposal_response$ from the endorsing node is not received after the set timer expires, you can choose to retransmit the $proposal$, which further reduces the probability of not generating an endorsement node.

(5) The sorting node listens and receives all transactions in the whole network, and packages the transactions into block $block < \{tx\}^m_{sig}$, $\{tx\}^m$ represents m ordered transactions contained in block. The sorting node signs the block and broadcasts it.

(6) After receiving the block, the submitting node first verifies the signature to check the integrity of the block, then verifies the read-write set, and updates the ledger accordingly. When each submitting node completes the above operations, it can be considered that the transaction initiated by the client has been reached consensus.

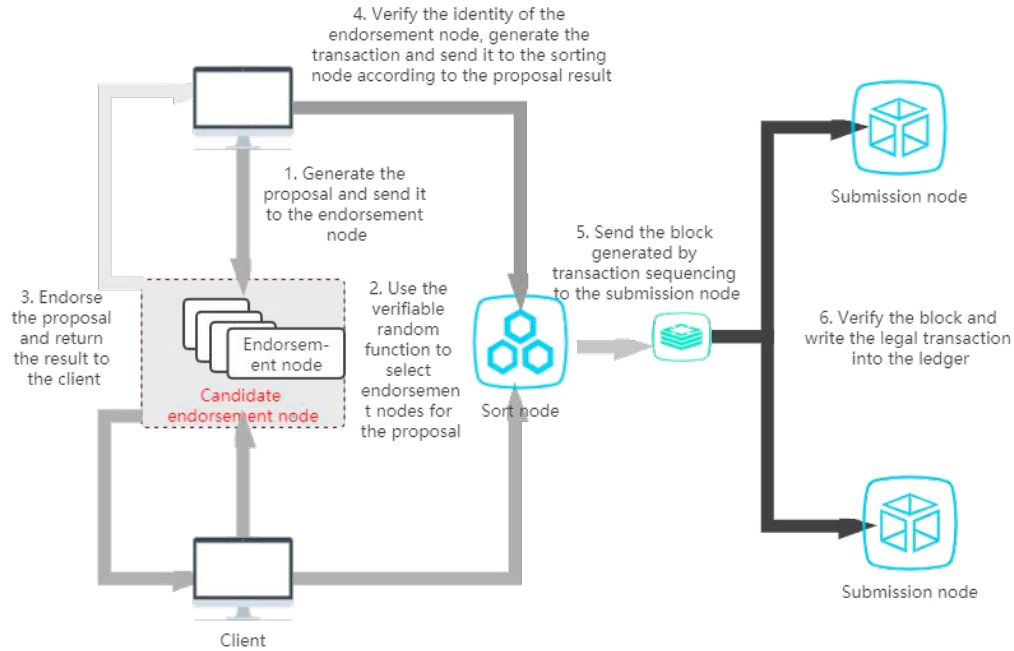


Fig. 2. The optimized consensus mechanism.

4. The experimental test

4.1. Experimental design

This paper constructs an electronic order system according to the Hyperledger Fabric architecture. The system simulates the electronic order process of Hyperledger Fabric. In particular, the system can select the endorsement nodes in fixed and random ways, that is, the system can conduct transactions according to the original scheme and the optimized scheme respectively. The experimental environment of this paper is a 12-core CPU, 16G memory, 2T mechanical hard disk server. The experiment USES Docker technology to deploy nodes on the server in the form of containers. There are four types of nodes in the system: client, candidate endorsement node, sorting node and submission node. The nodes are interconnected through the network, which can carry out broadcast communication and point-to-point communication. The node settings in this experiment are as follows: 100 clients, 10 candidate endorsement nodes, 1 sorting node, and 100 submission nodes. Clients are independent of each other and can submit electronic orders at the same time. When the system uses the original scheme, all candidate endorsement nodes need to endorse the electronic order. When using the optimized scheme, the extraction process of endorsement nodes will be increased (extraction threshold $\lambda=0.4$), and the client will also increase verification endorsement nodes accordingly. The process of identity. In addition, the ordering and submission strategies of the original plan and the optimized plan are consistent. Through the performance test comparison of the two electronic order submission schemes, this article will analyze the optimization scheme in terms of security and transaction processing speed.

4.2. Security analysis

When there are 10 candidate endorsement nodes ($n=10$) and the threshold $\lambda=0.4$, the probability of the adversary's successful attack is shown in Figure 3:

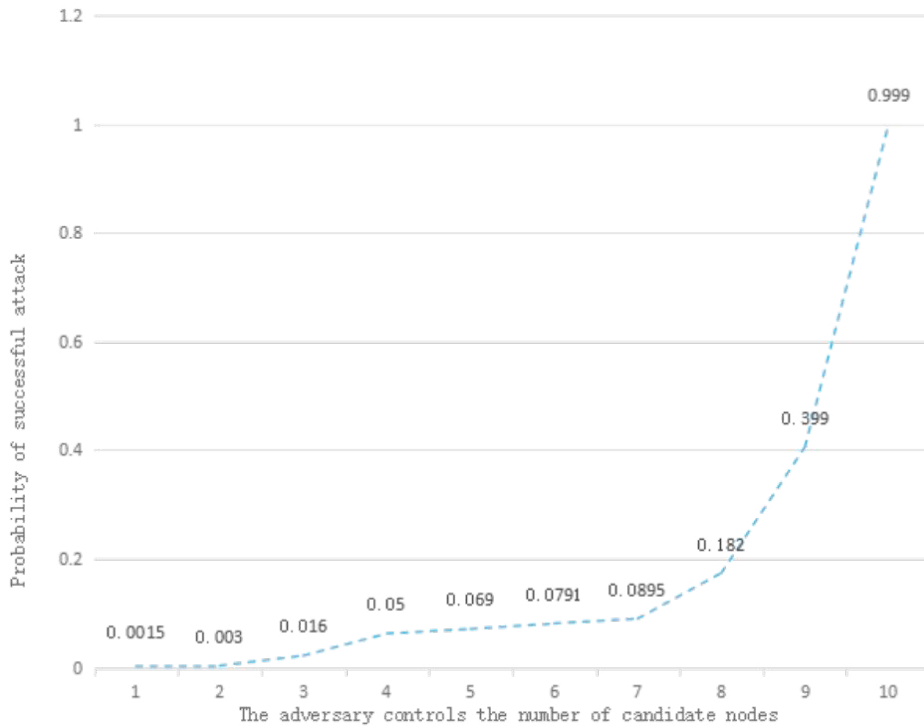


Fig. 3. Probability of successful enemy attack consensus mechanism

It can be seen from Figure 3 that the probability of an adversary's successful attack will increase with the increase in the number of control candidate endorsement nodes, and the trend of change increases exponentially. Although when the adversary controls all candidate endorsement nodes, the probability of successful attack will approach 100% (There may not be endorsement nodes, so the probability is not 100%), but when the adversary does not fully control the endorsement candidate nodes, it can still guarantee that the adversary will attack successfully with a lower probability, so it can be passed Set λ and n (the number of candidate endorsement nodes) to meet different security requirements.

4.3. Electronic order processing speed analysis

The experiment respectively simulated the original Hyperledger Fabric consensus mechanism and the optimized consensus mechanism. The transaction system of the two schemes was constructed through simulation, and the transaction processing speed of the two schemes was compared. The system parameter setting is: endorsement The number of node candidate sets is 10, and the threshold λ is 0.4. The experimental results are shown in Figure 4.

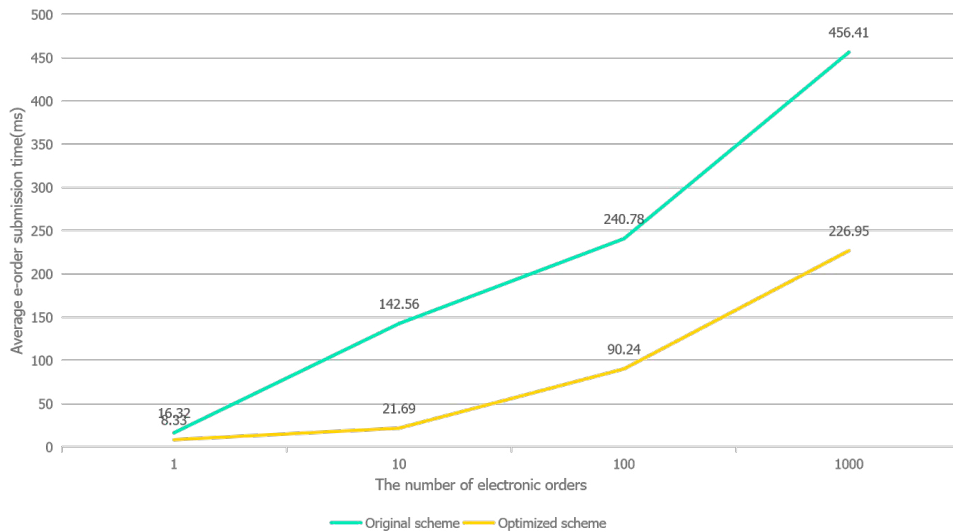


Fig. 4. The comparison of transaction time between original scheme and optimized scheme

As can be seen from the experimental results in Figure 4, compared with the original scheme, the electronic order processing speed of the optimized scheme has been significantly improved. Especially when the number of electronic orders is large, although the transaction time of both shows an upward trend with the increase of the number of electronic orders, the optimized scheme can still reduce the time of the original scheme by about 50%, and the processing speed of the optimized scheme is significantly improved.

5. Conclusion

In this paper, aiming at the existing problems of the original consensus mechanism of Hyperledger Fabric, an optimization scheme based on the endorsement node of verifiable random function is proposed, and the optimized consensus mechanism is analyzed and compared from the aspects of security and performance, and the experimental test is carried out through the electronic order system. The experimental results show that the optimized consensus mechanism has higher security, lower transaction delay and faster transaction processing speed on the premise of keeping the communication cost basically unchanged.

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